

CSCI 150 – Lab 3  
Due: Wednesday, Week7  
Submit to: Moodle

**Objectives:**

- Examine the Combinational **Design** Procedure
- Review the **Karnaugh Map** Reduction Method
- **Build circuits** to verify combinational design procedures
- Design and implement different decoders.

**Lab activities:**

1. Login into your Columbia College account
2. Right click on Start button, open Windows Explore
3. Create a new sub folder called **Lab3\_ yourFirstName** under your One Drive.
4. Read the follow the lab instructions and complete the circuit design by steps.
5. Save your practice to the file called **Lab3\_1.cct, Lab3\_2.cct**, under your Lab3 folder.
6. Observe the value changes at every I/O signal checkpoint (**Simulation /Show Value**)
7. Verify the circuits with **Truth tables**.
8. For each circuit diagram, write your Name, and the function Boolean equations beside the diagram
9. Save **screenshots, truth tables, K-map, function equations, etc..** in **Lab3\_report.doc** file.
10. Note: **Using LaTeX** for the Algebra expression whenever there is one to write.

## Combinational Design Procedure:

The steps taken for a combinational design are as follows:

- 1) Define/describe/state the problem (Specification)
- 2) Determine the number of variables needed and assign letter symbols to the input and output variables
- 3) Construct the truth table that defines the relationship between the inputs and outputs
- 4) Simplify the Boolean expression for each output, using K-Maps or Boolean algebra.
- 5) Draw the logic diagram
- 6) Implement the test circuit

## Lab Tasks:

### 1. Design a BCD-to-Seven Segment Decoder using NAND and NOT gates

#### Specification:

- Design a BCD-to-Seven Segment Decoder
- A 7-segment display can display the digits 0 - 9 by having the appropriate LEDs turned on.



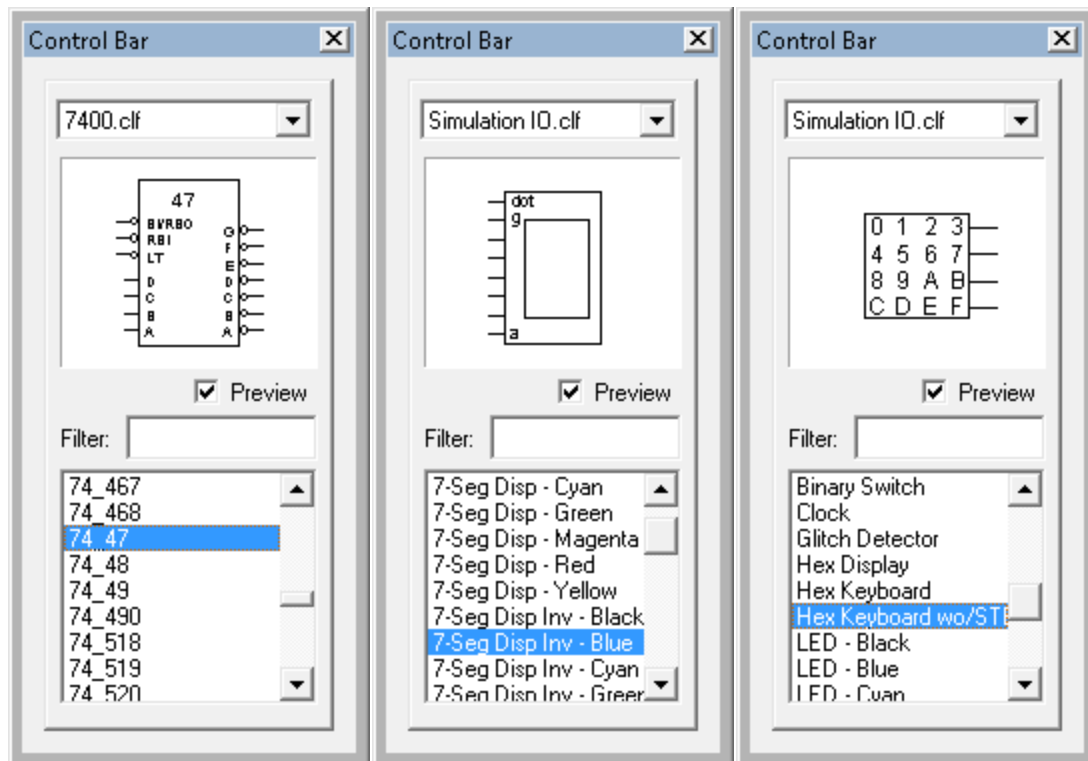
- For example, for the numbers 0, 3, and 4, segment "c" should be lit; for numbers 1, and 2, segment "c" should not be lit; etc.
- Design the circuit so that if someone applies a value from 10 to 15 then the LED segment does not light.
- You may choose to only use NAND, NOT and OR gates

**Tasks:**

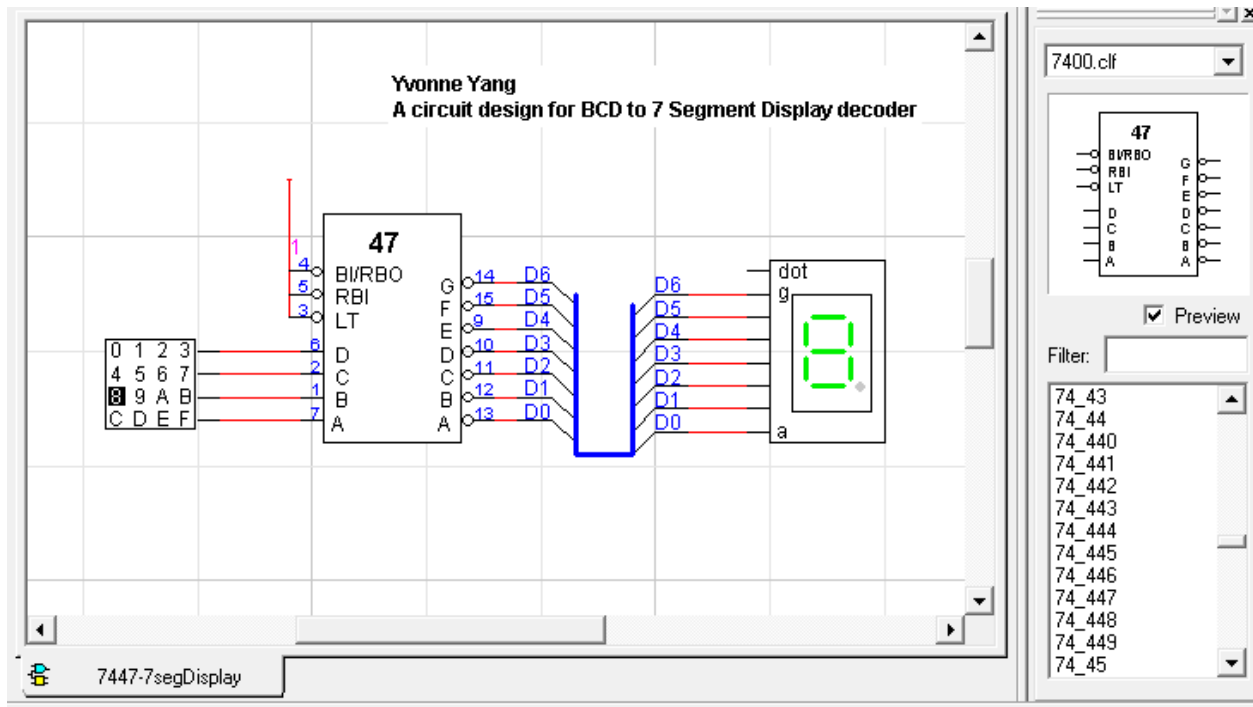
- 1) Determine input and output variables
- 2) Build the **truth table** to determine the Boolean function with **miniterms**
- 3) For each output (**a, b, c, d, e, f, g**), use **K-map** to simplify the Boolean function.
- 4) Draw the circuits and simulate the circuit in LogicWorks. Choose the **7-Segment Display** for outputs.
- 5) Choose **2 numbers from 0-9** as examples, show them with LogicWorks.
- 6) Test and make sure for **a number bigger than 9**, there will be **no output**.
- 7) Save your circuits as **Lab3\_1.cct**.
- 8) Save the above process of 6 steps in **Lab3\_report.doc**.

**2. Design a BCD-to-Seven Segment display using a build-in decoder****Tasks:**

- 1) See the sample diagram below, construct the circuit of BCD to 7 Segment Display, using a **decoder 7447 (from the 7400 Library)**. And use a **Hex Keyboard wo/STB** for inputs, and a **7-segment display** for outputs.
- 2) From LogicWorks, chose parts: **“74-47” decoder (with inverted outputs)**, **“7-Seg Disp Inv” (with inverted inputs)**, and **“Hex Keyboard wo/ST”**.



- 3) You can use a **Bus** to wire the signals between the decoder and the 7 segment display device. (Edit => Draw Bus). Create two **Breakouts** to connect the two devices: right clicking on the Bus, select **Breakout..**, then enter **D0..7**
- 4) Verify if each decimal number is displayed correctly.
- 5) **Bonus:** How to **disable** the inputs for the number of **10-15**?
- 6) Save the circuits in **Lab3\_2.ctt**
- 7) Save the screenshot in **Lab3\_report.doc**



### 3. Submit your work:

- Make sure you have created the following compressed file including file:
  - Lab3\_yourID\_yourName.zip (compressed file)
    - ✓ Lab3\_1.cct
    - ✓ Lab3\_2.cct
    - ✓ Lab3\_report.docx (truth tables, formulations, K-maps, screen shots)
- Submit the zip file to Moodle